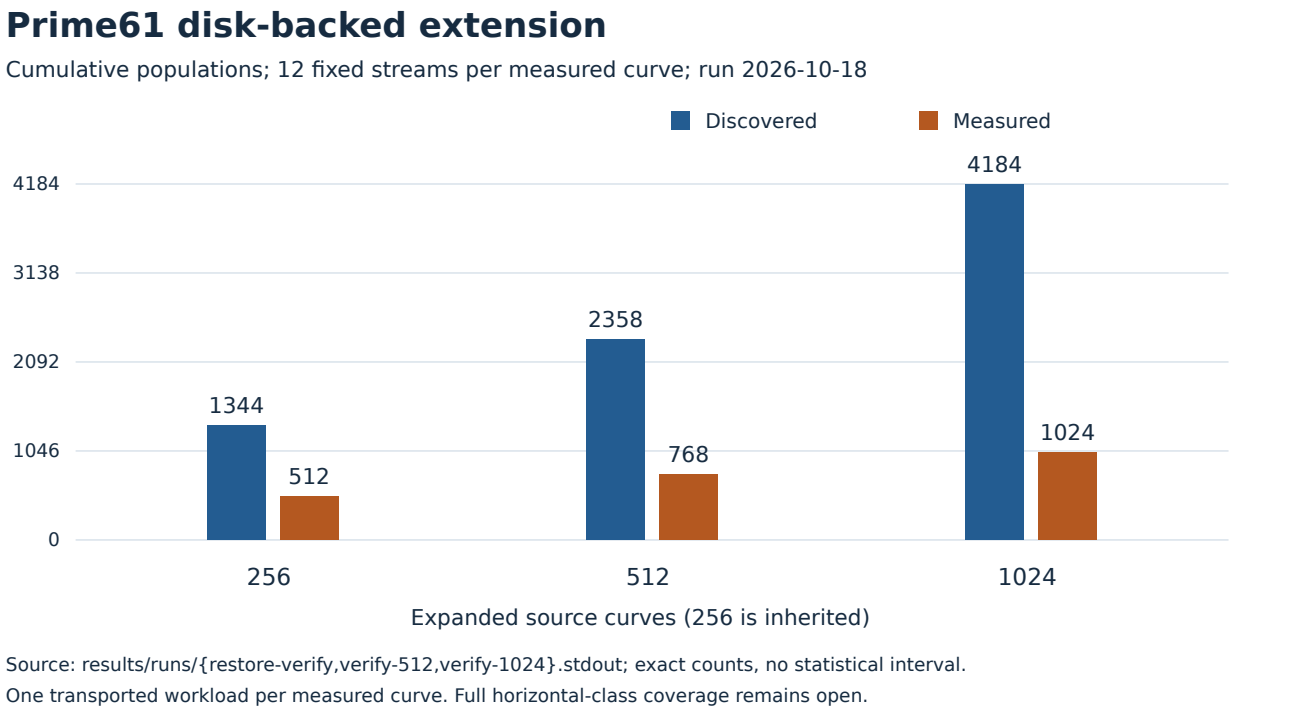
Prime61: larger checkpointed experiment

2026-10-18 · native capacity and capped relation-stage observations

The fixed public synthetic experiment over $p = 2^{61} - 1$ continues from the verified 256-source checkpoint. Counts below are cumulative. The first row is inherited; all earlier evidence remains frozen. Every measured curve receives twelve seeded streams with the same collector limits.

Expanded	Discovered	Maps	Measured	Streams
256	1344	3840	512	6144
512	2358	7680	768	9216
1024	4184	15360	1024	12288



Source: results/runs/{restore-verify,verify-512,verify-1024}.stdout; exact counts, no statistical interval.
One transported workload per measured curve. Full horizontal-class coverage remains open.

The final committed population has 1228800000 target trials, 0 relations and 0 verified solves. New since the input checkpoint: 2840 discovered curves, 11520 maps, 6144 streams and 614400000 target trials. The separate 192-stream collector replay is a control, excluded from these totals.

The live database occupies 11173888 bytes. The maximum reported native-worker peak RSS in the phase receipts is 20044 KiB. Compiler, test, native audit, Lean and renderer memory are excluded from this worker statistic. These are observed values on this prefix, with no full-class resource extrapolation or admitted wall-speed comparison. Full IC cost, S and speedup remain unknown.

There are 3160 discovered sources pending. The selected-prime component remains open. Neither this prefix nor its successful checks establish a class-wide factor-base ranking or completed enumeration.

Workload, controls and durable state

Worker source is pinned to `b68341cd72b51acf9c5ce6c4f755351cc83053db`, with Rust 1.99.1 and system SQLite. This is the same validated source as the input checkpoint. The new protocol was frozen before extension; unrelated current-main compilation failures remain recorded in the preceding study.

The unchanged spec fixes $a = p - 3$, $b = 111$, prime subgroup order `2305843010818082053`, `G=(1,1796364306280342617)` and `Q=(992726311114892094,558260110083040833)`. The checking scalar 539365399 is public synthetic data; the collector receives G and Q. Degrees are 3,7,11,13,19,23,53,61. Each curve uses a fresh low-x base with three inverse pairs, twenty requested relations, and seeds 2026101600 through 2026101611, capped at 100000 target trials per stream. There is one first-parent transported workload per curve.

The inherited snapshot was preserved locally from the previous verified Drive round trip. Before expansion, the native verifier checks every inherited map and source roster and replays 192 collector streams; its logical dump must match exactly. Every batch then checks all maps, expanded-source rosters, points, relations, scalars and run flags. The separate native prefix audit compares all 11330 inherited immutable rows; only expanded/measured flags may advance. Raw commands, outputs, errors and exits, including any caps, remain in `results/runs.tar.xz`.

Expansion chunks have at most 32 sources; collection chunks at most 64 curve IDs, each with a 300-second outer limit. Full verification has a 900-second limit for its larger checked population. A timed-out transactional phase may resume against committed state with a new log; no stream limit is reduced. The worker uses an 8 MiB SQLite cache, disabled mmap and disk-backed temporary storage. Disk data still grow.

Both closed batch snapshots are packaged as immutable Drive archives, metadata-read back, downloaded, manifest-checked and compared byte for byte. The final downloaded database produces the same native logical dump. Sanitized receipts omit private Drive identifiers; small archive round trips do not establish full-class quota. No remote compute or post-session experiment is provisioned.

Pinned-source library/example tests, native diagnostic tests, Lean build/axiom checks, archive recovery and report inspection are recorded in the README and `results/`. Host isolation is unqualified: five visible Intel Xeon Platinum 8573C CPUs, four-CPU cgroup quota, 16 GiB memory. Canonical performance graphs remain unchanged; this study adds prefix/finite-proof evidence without an admitted performance result.

Sources: `PROTOCOL.json`, `spec.json`, `continue.sh`, native worker sources in the pinned commit, raw archive and compact verification receipts, `visual.rs`, `reports/summary.json`, and the study README.

Selected-prime coverage: what the diagnostic adds

Selected-prime genus-character audit

Exact finite calculation: rank 4; 16 of 16 even-parity patterns; run 2026-10-18

Degree	-3	97	113	-35831	-5643470711
3	-1	+1	-1	+1	+1
7	+1	-1	+1	+1	-1
11	-1	+1	+1	-1	+1
13	+1	-1	+1	+1	-1
19	+1	-1	-1	-1	-1
23	-1	-1	-1	+1	-1
53	-1	+1	+1	-1	+1
61	+1	+1	+1	-1	-1

Derived using standard genus theory and the prior exact order certificate:
lcm(16, 68,162,026) = 545,296,208 divides the generated subgroup size.

If the whole class number is 545,296,208, full generation follows.
That class number remains conditional; class-group and CM bridges await Lean formalization.

Sources: results/genus-span.json, CLASS-COVERAGE.md, proof-map.json; finite Lean audit: 11 theorems.
This diagram describes a quotient calculation and a conditional implication; enumeration remains incomplete.

The native character calculation checks primitive positive forms for the eight selected degrees and norms coprime to the discriminant. Independent elimination and enumeration of all 256 subsets agree. Exact form data, 192 additional primitive representations and square-set controls are retained in `results/genus-span.json` and native test logs.

The subgroup lower bound uses standard genus theory and the preceding exact order-68162026 class certificate. It is a mathematical derivation, not an enumerated vertex count. Full generation follows **if** the entire class number equals 545296208; that number remains conditional after the earlier certification timeout. No class-number upper bound is proved.

Lean 4.34.1 checks eleven finite/arithmetic theorems, including complete XOR coverage, coefficient identities, the LCM and conditional size implications. The repository axiom audit passes with zero extra trusted axioms; standard foundations are recorded. Formalization is partial: norm-to-character, ideal-class/genus, the old power certificate and CM action bridges remain explicit obligations in `proof-map.json`.

The companion coefficient identity is `IDC1h5d70c781a75cea1c`, with a pointer to the universal integer Lean proof and 32+32 native evaluation checks over $\mathbb{F}_{2^{61}-1}$. It does not certify class coverage. The existing fundamental-discriminant result continues to exclude vertical base-field edges for this fixed class.

PR 2144: study, immutable evidence, proof map and remaining obligations

Exact selected component: additive certificate

Exact selected-subgroup certificate

H = <G> × <T0,T1,T2,T3> ≅ C34081013 × C2^4

|H| = 34,081,013 × 16 = 545,296,208

Degree	Exponent of G	Torsion mask
3	0	1
7	17040507	2
11	24833214	4
13	17619985	2
19	31141112	7
23	734461	3
53	2966921	4
61	22645521	8

G = A7²; T = [A3, A7^34081013, A11^34081013, A61^34081013].

All 16 torsion products distinct; eight exact relations checked without a class-number oracle.

Standard proper-class-group and CM bridges are outside the partial Lean proof.
Whole-class equality remains uncertified. The experiment has enumerated a prefix.

Sources: results/class-relations.jsonl, verify_relations.gp, COMPONENT-CERTIFICATE.md.
Finite coordinate bijection: ComponentCoordinates.lean. Run 2026-10-18; exact witnesses, no interval.

The odd generator has exact prime order 34081013. Each torsion basis element squares to the principal form and all sixteen subset products are distinct. Odd-order and exponent-two subgroups intersect trivially. The explicit relations show containment in both directions, giving exactly 545296208 selected ideal classes. Standard ordinary maximal-order CM theory transfers this size to the selected-prime curve component. This is a mathematical certificate, separate from actual enumeration.

The frozen producer uses exact PARI 2.15.2 reduced-form arithmetic and a 5838-entry baby/giant table. A separate fixed-witness checker performs no logarithm search and rejects both a corrupted exponent and a corrupted torsion mask. Both paths use the same PARI library and host. No whole class-number oracle is used. The prior power certificate also replays byte identically.

The equivalent structure `C68162026 × C2^3` suggests unique coordinates for a future streaming backend. Lean proves the finite encode/decode bijection and count. The complete project has fourteen authored theorems; the gate audits sixteen including two generated equation lemmas, with zero extra trusted axioms. **Formalization remains partial:** the native class arithmetic, group interpretation, cardinality argument and CM bridges remain outside Lean. See `component-proof-map.json`.

This sharpens the preceding genus lower bound without changing its frozen evidence. It supplies no whole-class upper bound and completes no traversal. An index-free coordinate walker has not been implemented or tested; this run still uses SQLite BFS and a growing disk index.

Sources: `RELATIONS_PROTOCOL.json`, `class_relations.gp`, `verify_relations.gp`, `results/class-relations.jsonl`, `results/class-relations-witness-check.json`, `COMPONENT-CERTIFICATE.md`, `component_visual.rs`, `proof/ComponentCoordinates.lean` and `results/component-proof-gate/`.